Mayur G D

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Summary

Data Scientist & ML Engineer (5+ yrs) specializing in large-scale forecasting, MLOps and high-performance ML systems with experience in Agentic AI and Generative AI technologies. Proven impact across retail, finance and enterprise platforms with strong experience in optimization, cloud infrastructure and LLM-based automation.

Experience

Deloitte USI - Data Scientist (Aug 2020 – Present)

MLOps Platform Onboarding Assistant [Agentic AI, LLMOps] *[Aug 2025 – Mar 2026]*

- Reduced MLOps use case onboarding from **3–5 days to <60 minutes** by automating GitHub repo creation, Databricks Unity Catalog schema + compute provisioning via **conversational AI interface**
- Built stateful multi-turn **agentic backend** on AKS using Microsoft Agent Framework with **thread serialization** per session and **Redis-backed session store** for production multi-user concurrency
- Implemented **RAG pipeline** using **ChromaDB + custom embeddings** with incremental MD5 checksum indexing on wiki docs, served via dedicated **FastAPI** knowledge base api
- Deployed **self-hosted Langfuse observability stack** on **OpenShift** via custom Helm charts with **Entra ID SSO** and **Azure Key Vault** secrets with dedicated PostgreSQL, Redis, Blob Storage resources created on Azure
- Designed **central MCP server** exposing standardized tool-calling interface across all enterprise agents, automated CI/CD via **Docker, Helm, and GitHub Actions**

Demand Planning Inference Optimization [MLOps]: *[June 2025 – July 2025]*

- Led onboarding and optimization of **XGBoost pipeline**, rewrote key components in PySpark-SQL + caching, reducing inference runtime 17.5hrs → 33min **(97% reduction)** enabling SLA-compliant runtimes

Airlines Agent Support Platform [Gen AI] *[Feb 2025 – May 2025]*

- Built a real-time LLM support agent with Kafka-based streaming, topic detection across 42 airline categories, and RAG retrieval using MongoDB Atlas vector store, reducing **resolution time to <10 seconds**
- Implemented with OpenAI embeddings, Fusion Retrieval using BM25 and semantic re-ranking to surface contextualised **knowledge cards** directly into call-centre tooling in near real-time
- Designed a thread-safe concurrent session manager with conversation history, enabling stateful multi-turn RAG interactions

Financial Analytics Assistant [Agentic AI]: Developed a **conversational BI agent** enabling CFOs to query financial data using natural language; implemented NL2SQL generation, validation loops, and automated executive report generation

Pickup Order & Labour Forecasting [Time Series Forecasting] *[July 2023 – Jan 2025]*

- Led end-to-end design of an order & item forecasting system achieving **<1 order/hour MAE** across 2,500 stores, powering a **real-time task allocation** system for associates
- Established **model selection pipeline** by benchmarking naive baselines, windowed moving averages, ARIMA/SARIMA, and deep learning models - selecting optimal approach by predictive performance and scalability
- Eliminated PubSub-CloudFunction hops cutting execution latency from **2s to 300ms** using direct Cloud Run API calls, enabling staffing optimisations delivering **$17M+ in annual savings**
- Built **alpha/beta staged rollout** infrastructure with API-level endpoint versioning and **A/B testing** before full production release, enabling safe incremental deployment across **2,500 stores**

Journal Entry Analysis [ML Supervised & Unsupervised] *[Nov 2020 – Jun 2023]*

- Built unsupervised pipeline using PCA/UMAP/TSNE + HDBSCAN/GMM/KMeans with explainability via Kruskal + Decision Trees to analyze **millions of journal entries** across multiple financial service lines
- Delivered large-scale performance improvements across the **clustering pipeline** with up to **75% runtime reduction** using **PySpark-SQL** rewrites and **GPU-accelerated RAPIDS** models
- Deployed pipeline on Azure Databricks with automated job scheduling, **MLflow experiment tracking**, and structured logging enabling reproducible runs, **model versioning**, and rapid debugging across large-scale clustering workflows

Skills

Programming Languages	Python, SQL, PySpark
Cloud & Infrastructure	GCP, Databricks, Azure, Kubernetes, OpenShift, VertexAI, AWS
AI/ML	Supervised & Unsupervised, Deep Learning, Time Series, NLP, GenAI, AgenticAI, RAG, LLM
Frameworks & Libraries	FastAPI, Langfuse, MLFlow, PyTorch, TensorFlow
Agent Frameworks	LangChain, LangGraph, CrewAI, MicrosoftAgentFramework
Data & DevOps Tools	Docker, Redis, GitHub Actions, Helm, Argo, OpenShift, Backstage

Awards & Certifications

Databricks Certified Generative AI Engineer Associate [Dec 2025 – Dec 2027]
Databricks Certified Machine Learning Associate [Feb 2026 – Feb 2028]
4 Deloitte Performance Awards [2022, 2023, 2024, 2025]

Education

PES UNIVERSITY	**[2016 - 2020]**	B. Tech ECE.	9.24 GPA
JAIN COLLEGE	**[2014 - 2016]**	Pre-Uni Course – PCME	95.83 %
SADHGURU SAINATH SCHOOL	**[2014]**	Secondary School	10 GPA